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CSA0911

JAVA

1. Try – Catch – Finally Basks

**SIMATS**
ENGINEERING

SIMATS Online Programming Lab
JAVA PROGRAMMING

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QUESTIONS**Q1**
Try-Catch-Finally Basics
25 marks**Q2**
Producer-Consumer with
BlockingQueue
25 marks**Q3**
Synchronized Methods and
Race Conditions
25 marks

3/3 answered

Q1 Try-Catch-Finally Basics 25 marks

```
public class TryCatchFinallyDemo {
    public static void main(String[] args) {
        int[] numbers = {1, 2, 3};

        try {
            System.out.println(numbers[5]); // ArrayIndexOutOfBoundsException expected here
        } catch (NullPointerException e) { // Bug 1: wrong exception type caught, real exception not handled
            System.out.println("Caught null pointer");
        }

        try {
            int value = Integer.parseInt("abc");
        } catch (Exception e) {
        }

        String str = null;
        try {
            System.out.println(str.length());
        } catch (NullPointerException e) {
            return;
        } finally {
            System.out.println("Cleanup");
            str.length();
        }

        try {
            riskyMethod();
        } catch (IOException e) {
            e.printStackTrace();
        }
    }

    static void riskyMethod() throws java.io.IOException {
        throw new java.io.IOException("failure");
    }
}
```

Your Code

```
1 public class TryCatchFinallyDemo {
2     public static void main(String[] args) {
3
4         int[] numbers = {1, 2, 3};
5         try {
6             System.out.println(numbers[5]);
7         } catch (ArrayIndexOutOfBoundsException e) {
8             System.out.println("Caught array index error");
9         }
10        try {
11            int value = Integer.parseInt("abc");
12        } catch (Exception e) {
13            System.out.println("Caught number format error");
14        }
15    }
16 }
```

Input

stdin input

OutputCaught array index error
Caught number format error
Caught null pointer
Cleanup
Caught IOException: failure

2. Producer – Consumer with BlockingQueue

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QUESTIONS

Q1
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25 marks

Q2
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BlockingQueue
25 marks

Q3
Synchronized Methods and
Race Conditions
25 marks

3/3 answered

Q2 Producer-Consumer with BlockingQueue 25 marks

```
import java.util.concurrent.*;

public class ProducerConsumerDemo {
    public static void main(String[] args) {
        BlockingQueue<Integer> queue = new ArrayBlockingQueue<>(5);

        Thread producer = new Thread() -> {
            for (int i = 0; i < 10; i++) {
                queue.add(i);
            }
        };

        Thread consumer = new Thread() -> {
            for (int i = 0; i < 10; i++) {
                int val = queue.poll();
                System.out.println("Consumed: " + val);
            }
        };

        producer.start();
        consumer.start();

        producer.join();
        consumer.join();

        System.out.println("Queue size: " + queue.size());
    }
}
```

Your Code

```
1 import java.util.concurrent.*;
2
3 public class ProducerConsumerDemo {
4     public static void main(String[] args) throws InterruptedException
5
6         BlockingQueue<Integer> queue = new ArrayBlockingQueue<>(5);
7
8         Thread producer = new Thread() -> {
9             for (int i = 0; i < 10; i++) {
10                 try {
11                     queue.put(i);
12                     System.out.println("Produced: " + i);
13                 } catch (InterruptedException e) {
14                     Thread.currentThread().interrupt();
15                 }
16             }
17         };
18
19         Thread consumer = new Thread() -> {
20             for (int i = 0; i < 10; i++) {
21                 try {
22                     int val = queue.take();
23                     System.out.println("Consumed: " + val);
24                 } catch (InterruptedException e) {
```

Input

stdin input

Output

Consumed: 0
Produced: 0
Produced: 1
Consumed: 1
Consumed: 2
Produced: 2

3. Synchronized Methods and Race Conditions



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QUESTIONS

Q1
Try Catch Finally Basics
25 marks

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25 marks

Q3
Synchronized Methods and
Race Conditions
25 marks

3/3 answered

Q3 Synchronized Methods and Race Conditions25 marks

```
public class SynchronizedCounterDemo {  
    private int count = 0;  
  
    public void increment() {  
        count++;  
    }  
  
    public synchronized void decrement() {  
        count = count - 1;  
    }  
  
    public static void main(String[] args) throws InterruptedException {  
        SynchronizedCounterDemo demo = new SynchronizedCounterDemo();  
  
        Runnable task = () -> {  
            for (int i = 0; i < 1000; i++) {  
                demo.increment();  
            }  
        };  
  
        Thread t1 = new Thread(task);  
        Thread t2 = new Thread(task);  
        t1.start();  
        t2.start();  
  
        System.out.println("Final count: " + demo.count);  
  
        t1.join();  
        t2.join();  
  
        synchronized (demo.count) { lock object  
            System.out.println(demo.count);  
        }  
  
        int finalCount = demo.count;  
        System.out.println(finalCount);  
    }  
}
```

Your Code

```
1 public class SynchronisedCounterDemo {  
2     private int count = 0;  
3  
4     public synchronised void increment() {  
5         count++;  
6     }  
7  
8     public synchronised void decrement() {  
9         count--;  
10    }  
11  
12    public static void main(String[] args) throws InterruptedException  
13        SynchronisedCounterDemo demo = new SynchronisedCounterDemo();  
14    }
```

Input

stdin input

Output

Final count: 2000